Status	Finished
Started	Wednesday, 21 October 2026, 2:01 PM
Completed	Wednesday, 21 October 2026, 2:21 PM
Duration	20 mins 13 secs
Marks	14.67/15.00
Grade	**97.78** out of 100.00
Feedback	What A great effort!

Question 1

Complete

Mark 1.00 out of 1.00

An encoder which has 8 inputs and 3 outputs

Select one or more:

- [x] a. is a combinational circuit
- [x] b. has only one input = 1 at a time
- [ ] c. is a sequential circuit
- [ ] d. works for any 8-bit pattern on its inputs
- [ ] e. is specified as "3-to-8"
- [x] f. is specified as "8-to-3"

Question 2

Complete

Mark 1.00 out of 1.00

Is the following logical statement true or false? $\overline{\overline{ab}+c} = \overline{a}\overline{b}\overline{c}$

Select one:

- ○ True
- ◉ False

Question 3

Complete

Mark 1.00 out of 1.00

In the architecture of the ATmega32U2, the Status Register contains which of the following?

Select one or more:

- [x] a. An overflow flag bit
- [ ] b. An I/O enable bit
- [x] c. A zero flag bit
- [ ] d. An instruction flag bit
- [x] e. A carry flag bit
- [ ] f. A halt CPU bit
- [ ] g. An even flag bit

Question 4

Complete

Mark 1.00 out of 1.00

Which of the following statements are true for synchronous sequential circuits?

Select one or more:

- [x] a. They have feedback paths
- [ ] b. Inputs must change at the same time as the clock
- [ ] c. Every output is fed back to an input
- [x] d. They include finite state machines
- [ ] e. They must have the same number of inputs as outputs
- [ ] f. Their outputs depend only their inputs at the time of the clock

Question 5

Complete

Mark 1.00 out of 1.00

We would like to implement very accurate delays from 1 millisecond to 100 milliseconds using the ATMega32u2 microcontroller. We will use a 16-bit timer and a clock frequency of 8 MHz. What single prescaler value should be used in order to achieve the best precision over the range of delays?

Select one:

- a. clk(i/o) /1024
- b. clk(i/o) /256
- c. clk(i/o) /64
- d. clk(i/o) /8
- e. clk(i/o) /1

Question 6

Complete

Mark 1.00 out of 1.00



Which of the following statements are correct for the tristate buffer

Select one or more:

- [x] a. When the control line is 1, the output of the buffer has the same state as the input
- [ ] b. The name of the buffer is "tristate", because it works with trinary, not binary, logic.
- [ ] c. A tristate buffer is the same as a 2:1 multiplexer
- [ ] d. Every connection to a data bus must be via a tristate buffer
- [x] e. Buffers like this are used to connect the outputs of the general purpose registers to the data bus
- [x] f. When the control line is 0, the output of the buffer is OFF (high impedance)

Question 7

Complete

Mark 0.67 out of 1.00

Select all of the following that are correct in regards to the Control and Execute Unit of a microcontroller:

- [ ] a. It is a really, really big counter
- [x] b. It selects the operation of the ALU
- [ ] c. It directly performs arithmetic, branching, and data transfer operations
- [ ] d. It is effectively a 40:8 bus multiplexer
- [x] e. It is part of the CPU
- [x] f. it controls address buses
- [x] g. It directly manipulates the databus to ensure correct flow of data
- [x] h. It decodes the binary sequence within the instruction register
- [x] i. It manipulates and reads the special register file
- [ ] j. It is used to programme the microcontroller.

Question 8

Complete

Mark 1.00 out of 1.00

If you know that the 4-bit binary pattern 1010 represents a signed integer in 2's complement, which of the following decimal values does it represent?

Select one:

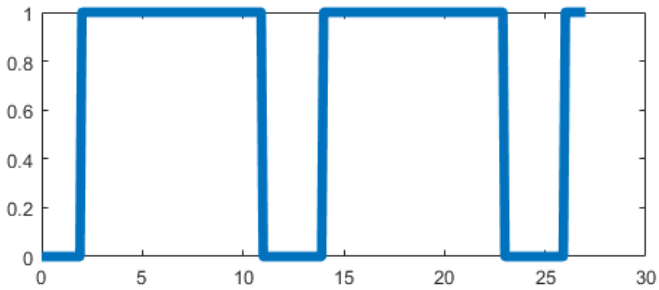
- a. -1
- b. 0
- c. 10
- d. 3
- e. -6

Question 9

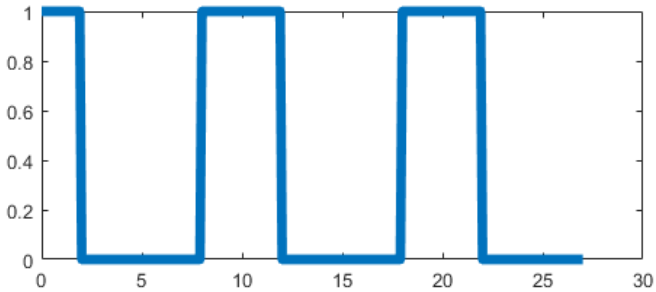
Complete

Mark 1.00 out of 1.00

A



B



The time axes in the plots are in milliseconds. Which of the following statements are correct for the two PWM waveforms shown?

Select one or more:

- a. The duty cycle for waveform B is 40%.
- b. The frequency for PWM waveform B is 250 Hz.
- c. The duty cycle for waveform A is 75%.
- d. The duty cycle for waveform B is 60%.
- e. The frequency of PWM waveform A is 120 Hz.
- f. The frequency of PWM waveform B is 100 Hz.
- g. The duty cycle for PWM waveform A is 25%.
- h. The duty cycle for waveform B is 67%.

Question 10

Complete

Mark 1.00 out of 1.00

Select all true options for the Stack memory of a microcontroller:

- [x] a. When information is written to the stack, the stack pointer decrements by one.
- [ ] b. The stack is used to store constant values that are referred to multiple times.
- [ ] c. When information is written to the stack, the stack pointer increments by one.
- [x] d. A 'stack frame' refers to all the data required to store the program context
- [x] e. The stack is used to store data that typically only needs to be read once.
- [x] f. The stack operates in a 'Last-in, First-out' style
- [ ] g. A 'stack frame' refers to the outside of the SRAM.
- [x] h. The stack pointer (SP) points to the address of the next free location on the stack.
- [ ] i. The stack pointer points to the address that has just had information sent to it.
- [ ] j. The stack operates in a 'First-in, First-out' style.

Question 11

Complete

Mark 1.00 out of 1.00

Within the ATMega32u2, the instruction register:

- [x] a. Contains 16 bits.
- [ ] b. Contains 8 bits.
- [ ] c. Directly selects the ALU operation.
- [x] d. Provides the opcode to the control/execute unit.
- [x] e. Stores the instruction about to be/being executed.
- [ ] f. Stores the result of the instruction.
- [x] g. Gets data from the program memory.
- [ ] h. Is connected to the same data bus as the EEPROM.

Question 12

Complete

Mark 1.00 out of 1.00

With respect to one of the **GPIO** pins on the ATmega32U2, **"pull up"** is

Select one or more:

- a. always connected
- b. used to ensure that the pin is read as LOW when nothing is externally connected
- c. able to be controlled by the program executing on the microcontroller
- d. used to ensure that the pin is read as HIGH when nothing is externally connected
- e. used to make sure the pin can be easily pulled out of a socket
- f. connecting a resistor between the pin and the supply voltage

Question 13

Complete

Mark 1.00 out of 1.00

The arithmetic logic unit (ALU) performs which of the following operations in a microprocessor?

Select one or more:

- a. pushing an operand onto the stack
- b. shifting one operand
- c. incrementing the program counter (PC)
- d. logical AND for two operands
- e. adding two operands
- f. outputting an operand to an I/O device
- g. compare for two operands
- h. loading an operand from memory

Question 14

Complete

Mark 1.00 out of 1.00

Select all true responses.

AVR instructions:

- a. Are an example of a RISC architecture.
- b. Are an example of a CISC architecture.
- c. Are stored as 0s and 1s.
- d. Can transfer data, perform calculations, and store data with a single instruction
- e. Are what the ATmega32u2 stores in its program memory.
- f. Are two bytes long.
- g. Are one byte long.

Question 15

Complete

Mark 1.00 out of 1.00

Syntax	Operands	Operation
`pop Rd`	$0 \leq d \leq 31$	Rd $\leftarrow$ STACK
Control registers	Opcode	Flags
PC $\leftarrow$ PC + 1 SP $\leftarrow$ SP + 1	1001 000d dddd 1111	None

Syntax	Operands	Operation
`push Rr`	$0 \leq r \leq 31$	STACK $\leftarrow$ Rr
Control registers	Opcode	Flags
PC $\leftarrow$ PC + 1 SP $\leftarrow$ SP - 1	1001 001d dddd 1111	None

Consider the hexadecimal sequence `0x923F`.

This hexadecimal sequence specifies the push instruction, which will transfer the contents of R3 to the stack .

◄ Quiz 3: Weeks 1-8

Jump to...

Lecture 1: Introduction ►

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